

SNAPFRAME · REQUIREMENTS

SnapFrame requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/snapframe		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

SnapFrame requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be checked by calculation at TRL 3 and revised after co-design sessions. The status column compares each target with the first-order estimates in SNF-PRC-001; every status is an estimate until TRL 3. Three requirements are **not met** by the reference concept: R6 (wind), R10 (cost) and R13 (fire), and R7 (skin coverage) is only partly met.

ID	Requirement	Target	Status (estimate)	Verification (TRL 3 or later)
R1	Give enough covered living space	Size M: 16 m ² floor or more, which is 3.5 m ² per person for four people (Sphere minimum)	Met: 4.0 x 4.0 m = 16 m ²	Geometry
R2	Give standing headroom	2.0 m or more over at least 60 % of the floor	Met: 2.0 m or more over the central 3.0 m of the 4.0 m span (75 %)	Geometry
R3	Cover several sizes from one design	At least three sizes (S about 9 m ² , M 16 m ² , L about 24 m ²) generated from one parametric node model; M and L share every tube length	Partly met: M modeled; S and L defined by parameters only	Parametric model generating all three node sets
R4	Go up without tools	Two adults with no tools and no prior training, using a picture guide, erect frame and skin of size M in 60 min or less	Estimate about 40 min; unverified	Time study in a later user trial (TRL 4 or later)
R5	Ship flat	No member longer than 2.1 m; all members straight; kit in two packages, each 25 kg (55 lb) or less and 0.10 m ³ or less	Met: longest member 2.06 m; tube bundle about 24 kg and 0.03 m ³ ; bag about 21 kg and 0.08 m ³	Massing model, then weighing
R6	Resist wind with the skin on	Every member stays elastic with a safety factor of 1.5 or more on yield in gusts of 20 m/s (72 km/h, 45 mph)	Not met: rafters about 1.26 with 3/4 in EMT; rating about 18 m/s (65 km/h, 41 mph)	Wind load calculation (CAL), then frame analysis
R7	Fit standard relief tarpaulins	Two 4 x 6 m tarpaulins (48 m ²) close the roof, both side walls and both gables	Partly met: full enclosure needs about 51 m ² plus overlaps; two tarpaulins close the roof, both side walls and one gable	Cutting and folding plan
R8	Anchor without a hammer	Every foot and guy anchored by hand; each anchor holds 1.0 kN uplift in firm soil	Unverified: estimated demand up to about 0.6 kN per foot anchor from wind, more with cable pretension	Anchor pull-out data for the chosen anchor and soils
R9	Keep nodes strong in sun and cold	Nodes carry the socket forces from R6 with a factor of 2 after two years outdoors, from -10 to 70 °C surface temperature	Unverified; node material not chosen	Material data, then node load tests (TRL 4)
R10	Stay within the concept budget	Complete size M kit \$400 or less in prototype quantities	Not met: about \$437 (9 % over); frame kit without tarpaulins about \$387	Priced BOM
R11	Be carried by two people	Complete size M kit 50 kg (110 lb) or less	Met: about 45 kg	Massing model, then weighing
R12	Be repairable in the field	Any member or node replaced by hand in 5 min or less; four tube types and three node types only	Met by design	Design review
R13	Limit fire spread			

ID	Requirement	Target	Status (estimate)	Verification (TRL 3 or later)
		Skin meets a recognized flame spread test for tents (for example CPAI-84) or the kit ships with a fire-retardant tarpaulin option	Not met with standard polyethylene relief tarpaulins, which are usually not flame retardant	Material data for the chosen tarpaulin

■ Assumptions

- Reference size M: 4.0 x 4.0 m floor, two 2.0 m bays, eave 1.8 m, ridge 2.6 m (roof pitch about 22°), three gable frames.
- Tube: 3/4 in EMT to ANSI C80.3 nominal dimensions (OD 23.4 mm, wall 1.24 mm), mass about 0.70 kg/m. Yield strength assumed 275 MPa (estimate, to be confirmed from supplier data).
- Wind: dynamic pressure $q = \frac{1}{2}\rho v^2$ with $\rho = 1.225 \text{ kg/m}^3$; windward wall coefficient +0.8; roof suction coefficient -0.7 (first-order values in the style of ASCE/SEI 7, not a code check). The tarpaulin panel is assumed to pass half its load to the frames and half to the eave and ridge tubes.
- Members are treated as simply supported between node centers; node fixity is ignored, which is conservative for bending.
- Snow is out of scope (SNF-PRB-001).
- Budget: \$400 is the `budget_usd` in `project.yaml`, applied here to one complete size M kit including tarpaulins, anchors and bags. Whether tarpaulins belong in the budget is proposed, awaiting Amish.